**Figure 1: In-PDF game execution.**

- ✓ **Real-time mouse interaction:** game engine response instantly to cursor movement, adjusting the paddle's position accordingly and ensuring precise and continuous interaction between player and game environment;
- ✓ **Dynamic scoring and visual feedback:** the score was continuously updated during gameplay, reinforcing player engagement and progression;
- ✓ **Automatic restart and status messages:** upon game over or victory, visual alerts were displayed, allowing immediate restart without reloading the document;
- ✓ **Partial cross-reader compatibility:** the game ran successfully in Google Chrome and Opera's built-in PDF viewers, maintaining all interactive features. However, incompatibilities were noted with Adobe Reader and other viewers, indicating the need for further adaptation to ensure broader support and consistent execution across platforms.

The complete source code, along with detailed usage instructions, is available at [OMITTED] as supplementary material, including the full engine implementation, an example game, and a demonstration video showcasing real-time gameplay and dynamic behavior within the PDF.

## 5 Conclusions and Future Work

This work presented an experimental study on the technical and conceptual feasibility of incorporating interactive and playful elements into digital documents, with emphasis on the PDF format. The investigation sought to understand the extent to which a traditionally static medium could be reinterpreted as a platform capable of hosting dynamic experiences. To this end, a functional prototype of a reusable game engine was designed and implemented, integrating game logic, simplified physics simulation, and real-time interaction handling, demonstrating the potential of PDF as an interactive environment.

Despite these achievements, the study also revealed limitations inherent to the PDF format, such as the absence of vector graphic APIs, restricted support for continuous user input events, and dependence on readers that correctly implement the JavaScript interpreter. However, these constraints do not undermine the experimental and conceptual value of the proposal, which demonstrates the capability of PDF to host interactive and playful applications without the need for external software or additional dependencies.

As future work, this research aims to extend the proposed game engine by enhancing its graphical and multimedia capabilities within the PDF environment, allowing the evaluation of its performance when handling complex visual resources and bringing the PDF-based experience closer to traditional digital games. The integration of sound effects, background music, and simple animations will also be explored to enrich player immersion while maintaining compliance with the security and rendering constraints of the format. Furthermore, the study intends to adopt and adapt the EnSCEvA configuration model (introduced in the EPUBForge project) as a standardized framework for defining game components, scenes, and behaviors, enabling the engine to interpret and execute EnSCEvA-described games directly inside PDFs. Building on this foundation, the research envisions the evolution of the current prototype into a comprehensive, document-oriented game engine capable of unifying the creation, execution, and distribution of interactive content across multiple digital document formats, promoting broader accessibility and reusability in educational, communicational, and experimental contexts.

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